

Searching for SUSY with delayed and non-pointing photons at ATLAS

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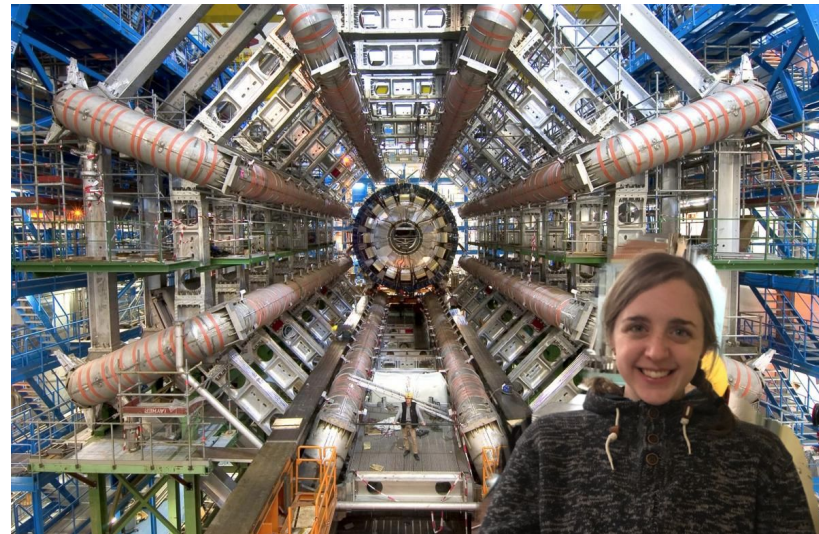
August 14, 2020



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Projects I've worked on within the NPP Analysis

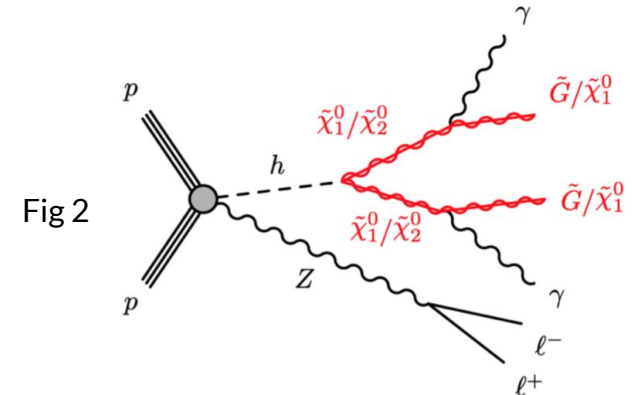
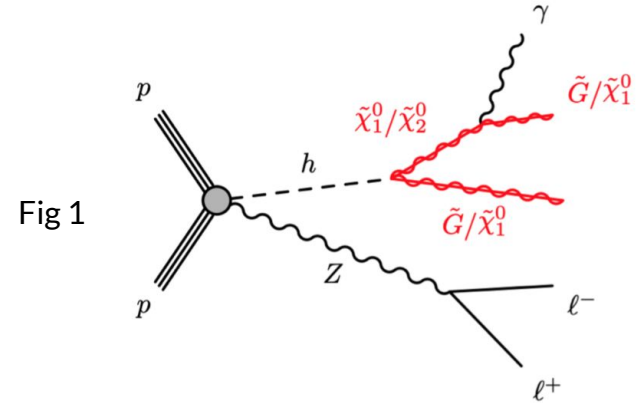
- Helped implement a photon timing correction to fix a negative timing bias in our data ✓
- Background templates: examining how photon purity affects distribution and statistics ✓
- Region defining: energy cuts to define the control region, validation region, and signal region



*Artistic depiction of what it would have looked like if I went to CERN this Summer

Supersymmetry (SUSY)

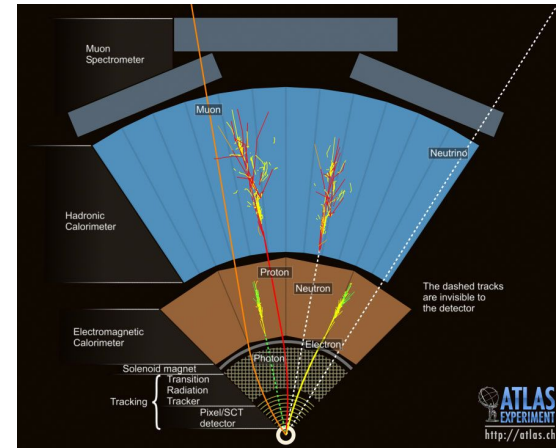
- SUSY is an extension of the SM where each SM particle has a corresponding SUSY particle
- It's a very appealing theory because it
 - Solves the Higgs mass problem
 - Provides candidate for dark matter particle
 - Unifies the strong, weak, and E&M forces during early universe $\rightarrow$ GUT
- GMSB (gauge-mediated symmetry breaking) theories predict the possibility of a Higgs decay to a long lived **NLSP** (next-to-lightest supersymmetric particle) which then further decays to an **LSP** (lightest supersymmetric particle) and a photon



Background Templates

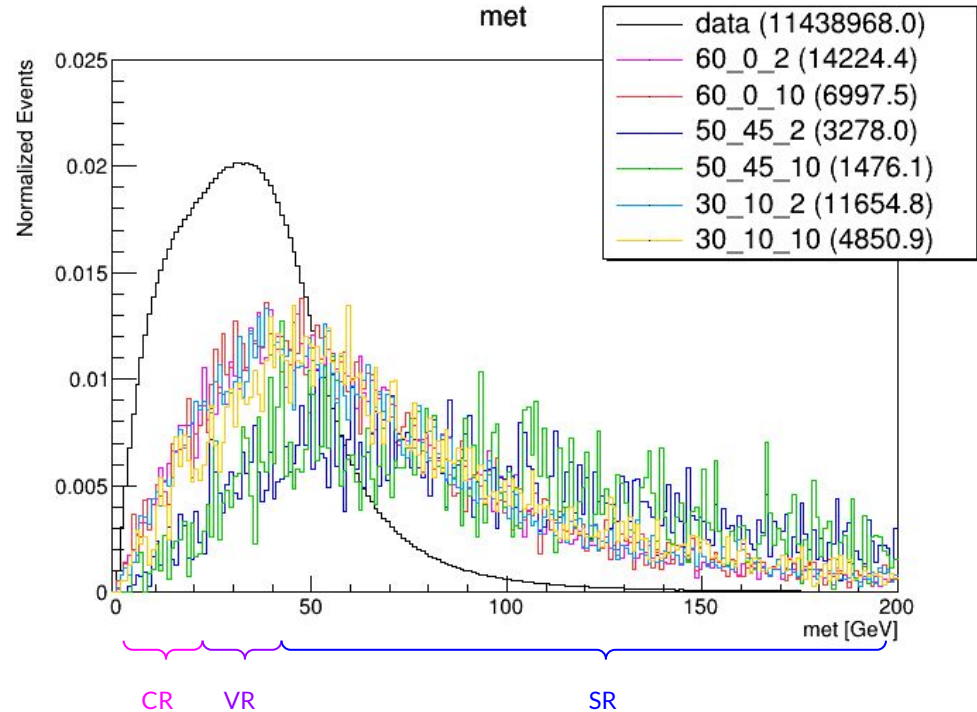
Detecting a signal depends on first being able to predict the background

- To do this, some portion of the data with a MET below a selected threshold is studied
 - MET < 20 GeV → Control Region
 - 20 < MET < 40 GeV → Validation Region
 - MET > 40 GeV → Signal Region
- Important note: the actual mass of the NLSP and LSP are left as free parameters in this study, so setting the MET threshold for the SR at 40 GeV is an educated guess
- We use the CR and the VR to make sure we understand our background
 - The detector picks up both real photons and objects resembling photons: $\ell\ell$, CR tight, and CR loose-not-tight
 - By modeling the distributions in timing and pointing of both samples, we will be able to approximate the VR and eventually the SR by a linear combination of the real and fake photon samples



MET cut for the SR

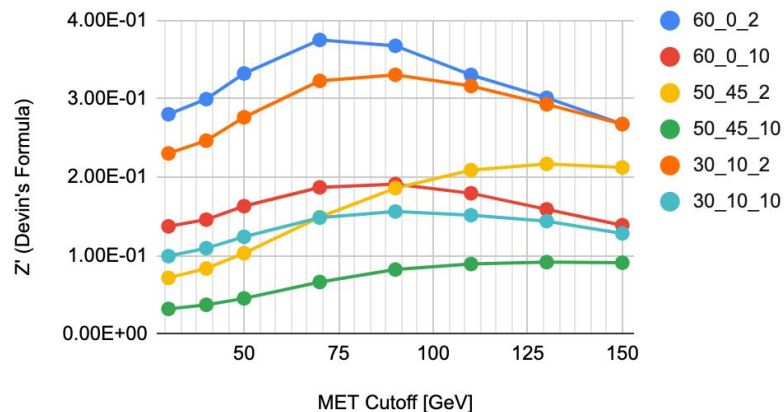
- The colored lines all represent different signal points simulated from the Monte Carlo
 - **60_0_2** → mass splitting 60/0, timing delay 2ns
 - The six MC signal points shown here have been selected from a much larger sample in order to show a range of possible mass splittings
- From the plot, we can see that the data begins to fall off around 40 GeV MET but that the signal remains relatively flat
 - Seems to indicate that 40 GeV is a good SR definition
 - Need to look at significance in signal to background to make sure



Inclusive channels

- We look at 6 signal points from the MC
- The significance peaks at a low of 70 GeV MET for the 60_0_2 signal point
- The peak in significance for all signal points is substantially greater than 40-50 GeV

Significance: All photon channels



MET Region	Data	60_0_2	60_0_10	50_45_2	50_45_10	30_10_2	30_10_10
S0 (> 30 GeV)		2.80E-01	1.37E-01	7.15E-02	3.19E-02	2.30E-01	9.93E-02
S1 (> 40 GeV)		3.00E-01	1.46E-01	8.35E-02	3.71E-02	2.47E-01	1.09E-01
S2 (> 50 GeV)		3.33E-01	1.63E-01	1.03E-01	4.53E-02	2.77E-01	1.24E-01
S3 (> 70 GeV)		3.75E-01	1.87E-01	1.49E-01	6.62E-02	3.23E-01	1.48E-01
S4 (> 90 GeV)		3.68E-01	1.91E-01	1.86E-01	8.21E-02	3.31E-01	1.56E-01
S5 (> 110 GeV)		3.31E-01	1.80E-01	2.09E-01	8.92E-02	3.17E-01	1.52E-01
S6 (> 130 GeV)		3.01E-01	1.59E-01	2.17E-01	9.16E-02	2.93E-01	1.44E-01
R9 (> 150 GeV)		2.68E-01	1.39E-01	2.13E-01	9.08E-02	2.68E-01	1.28E-01

Early indications on MET cut



- ❖ Early results suggested using a MET cut larger than 40 GeV
 - We need to look at pointing and timing statistics for more information

MET + Pointing & Timing

- ❖ Our first look at timing and pointing statistics introduced new questions and additional parameters to consider
 - Which binning to choose?
 - Which maxEcell cut off to choose?
 - How will the statistics in the background templates be affected?

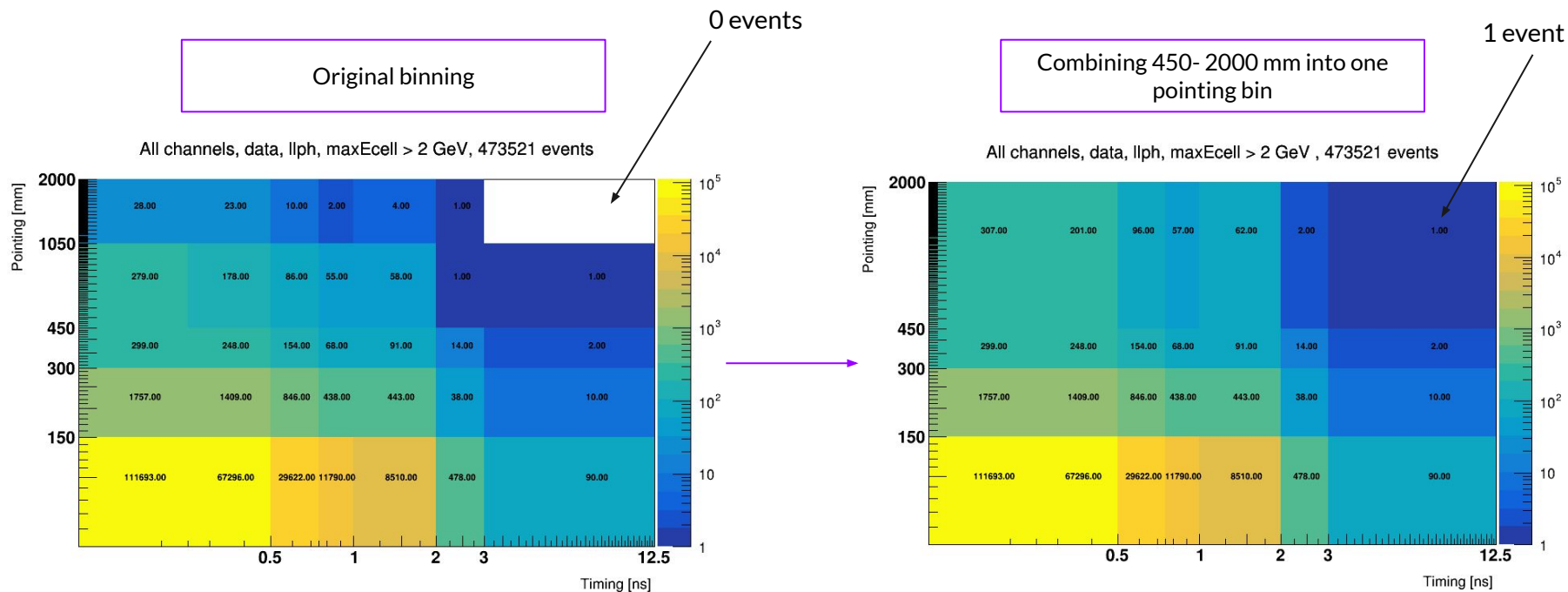
Binning Choice

- An optimization framework built by another member of our group suggested an **optimized** set of bins based on the use of a 5 GeV maxEcell cut
- We **adapted** this to make the bin spacing more uniform, monotonically increasing, and easier to justify
 - For timing, we use 4 bins < 1 ns instead of 5
 - For pointing, we use one fewer bin between 150 - 450 mm (inclusive channels), and two fewer between 0 - 150 mm (2ph channels)

	1ph Timing bins:	2ph Timing bins:
Optimized	[0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0, 3.0, 12.0]	[0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 12.0]
Adapted	[0, 0.25, 0.5, 0.75, 1.0, 2.0, 3.0, 12.50]	[0,0.25,0.5,0.75,1.0,12.5]
	1ph Pointing bins:	2ph Pointing bins:
Optimized	[0, 150, 200, 350, 450, 1050, 2000]	[0, 50, 100, 150, 350, 2000]
Adapted	[0, 150, 300, 450, 1050, 2000]	[0, 150, 300, 2000]

Binning Choices

- Early plots of **llph**, one of our background templates, suggested further adaptations to the binning for the inclusive channels



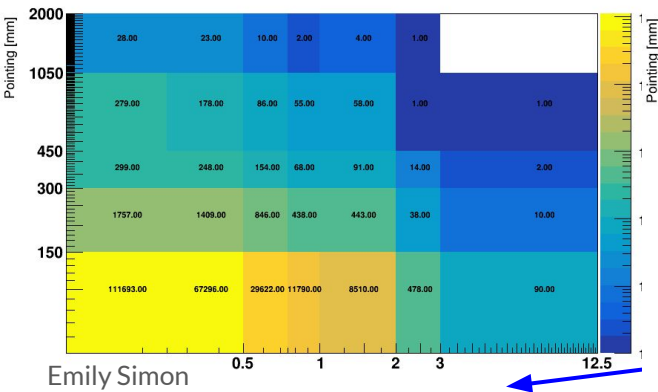
MaxEcell Choice

- Because changing the maxEcell cut can drastically change the statistics, we also needed to look in various maxEcell bins
 - 2, 5 and 8 GeV maxEcell cuts
 - 8 GeV does not look promising, 2 and 5 GeV are more realistic

2 GeV

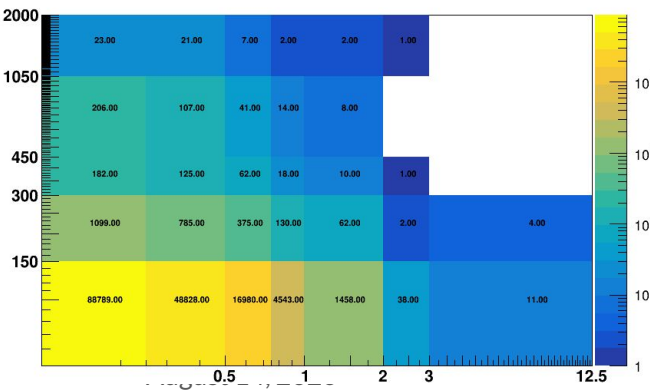
5 GeV

All channels, data, lph , maxEcell > 2 GeV, 473521 events



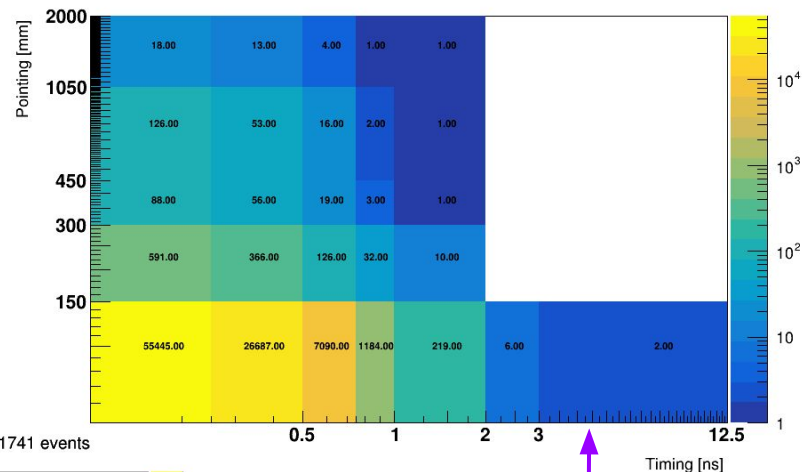
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All channels, data, lph , maxEcell > 5 GeV, 331741 events



8 GeV

All channels, data, lph , maxEcell > 8 GeV, 190040 events



Discouraging amount of empty bins

Empty bins could be corrected with reasonable changes to binning

Comparing Background Templates

Total Stats:

CR loose-not-tight:

756,925 events

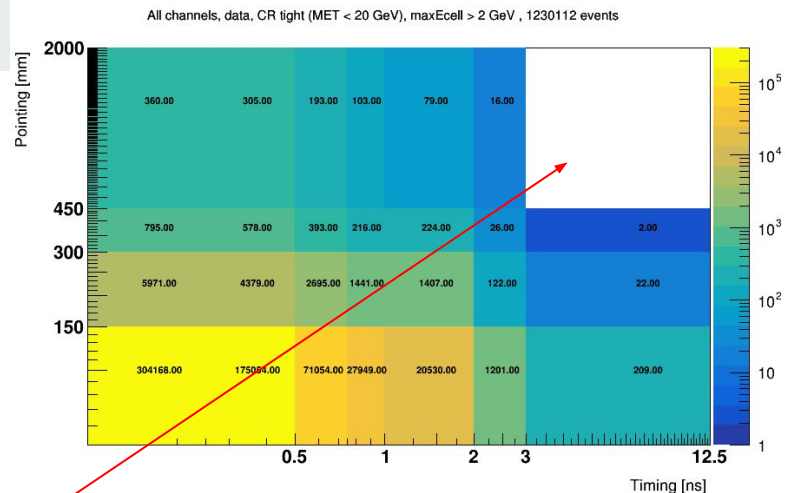
CR tight:

1,230,112 events

llg:

473,521 events

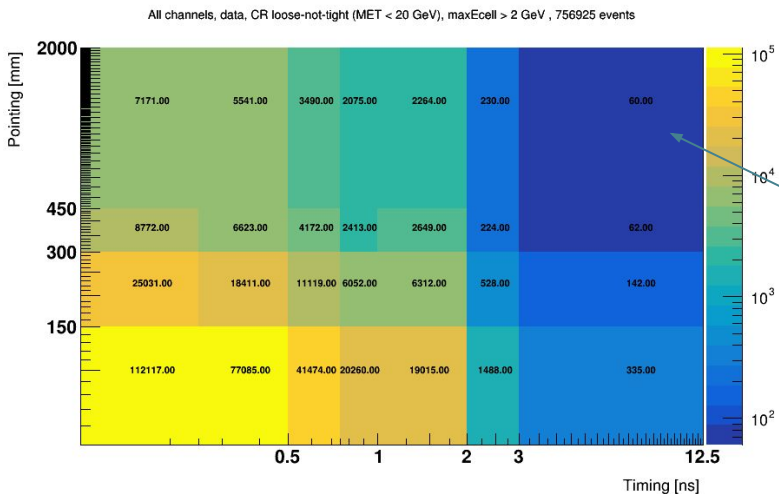
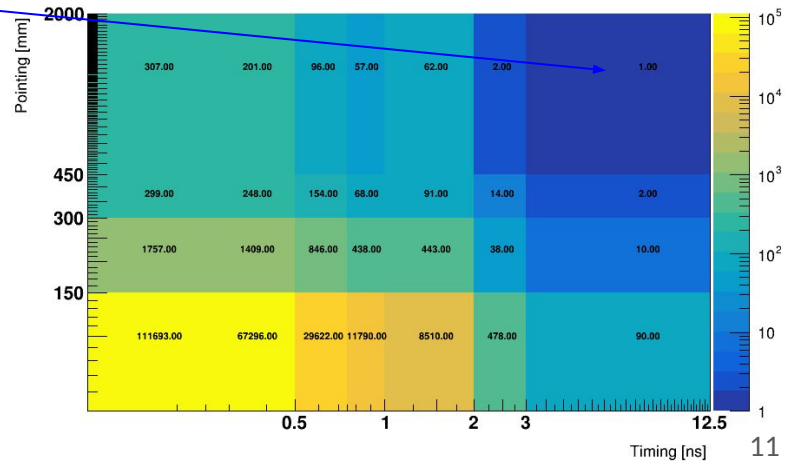
- ❖ Background templates are comprised of llg, CR tight, and CR loose-not-tight
- ❖ When all cuts are the same (MET, maxEcell):
 - We will be limited by real-enhanced photon template statistics, even though the total statistics are high, because the tails are much lighter.
- ❖ We've discussed combining CR tight and llg



Real photons:
CR tight: 0 events
llg: 1 event

CR loose-not-tight: 60 events

All channels, data, llph, maxEcell > 2 GeV, 473521 events



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Control Region Definition

- We explored increasing the CR MET threshold, since our earlier studies of the net significance suggested increasing the SR MET threshold will be favorable

CR MET < 20 GeV

All channels, data, CR tight (MET < 20 GeV), maxEcell > 2 GeV, 1230112 events

0 events

CR MET < 25 GeV

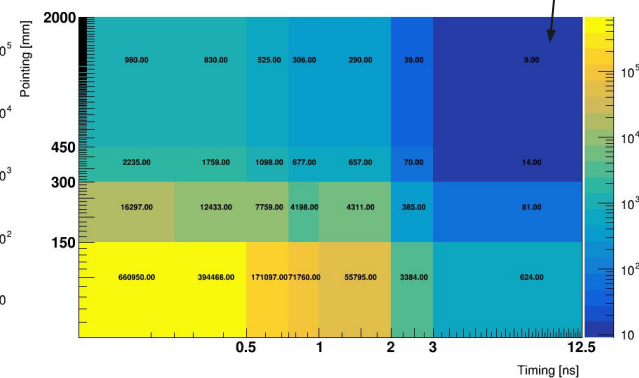
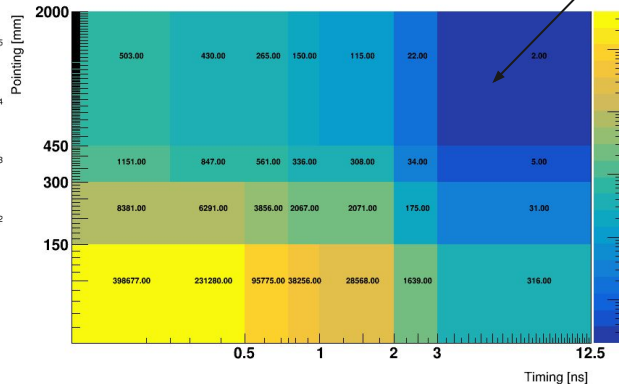
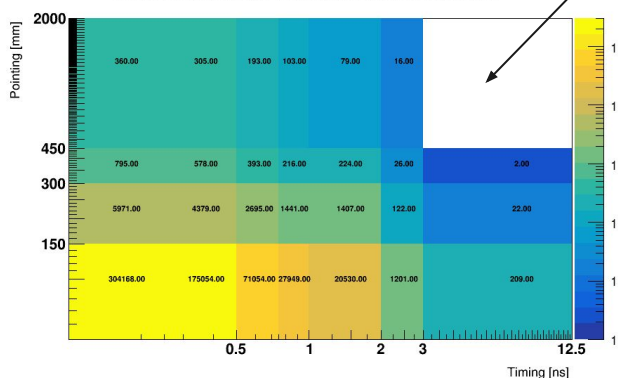
All channels, data, CR tight (MET < 25 GeV), maxEcell > 2 GeV, 1633329 events

2 events

CR + VR, MET < 40 GeV

All channels, data, tight (MET < 40 GeV), maxEcell > 2 GeV, 2807530 events

9 events



- Increasing the CR MET increases stats everywhere, and gives modest increases in the highest pointing and timing bins
- If SR MET is increased, we will revisit how to define the CR and the VR

Conclusions from the SR MET study

- ❖ Some key takeaways
 - Small adjustments in **binning choices** can help ensure we have enough events in important timing and pointing bins
 - Raising the **maxEcell** cut can greatly affect the number of statistics available in our background templates
 - To a certain extent, we can correct for this with coarser binning
 - Increasing the **MET** for the SR can give us way more events in the CR and VRs with modest increases in the extreme pointing and timing bins

- ❖ However, the interdependence of these parameters makes pinning down one optimal MET cut too difficult to do by eye
 - **Optimization** framework
 - With another member of the NPP team, I have made data covering all possible combinations of the search space which will be fed to Devin's framework

❖ *...Results soon to come!*

Proposed Search Space

Parameter	Allowed Values
$f_y (= 1 - f_j)$	0, 0.25, 0.5, 0.75, 1
maxEcell E cut [GeV]	2, 3, ..., 10
VR Low MET Cut [GeV]	30, 40
VR High MET Cut [GeV]	50, 60, 70, 80
$n_{\text{bins}, t}$	6-11
$n_{\text{bins}, z}$	1-6

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Julia Gonski
Kiley Kennedy
Devin Mahon

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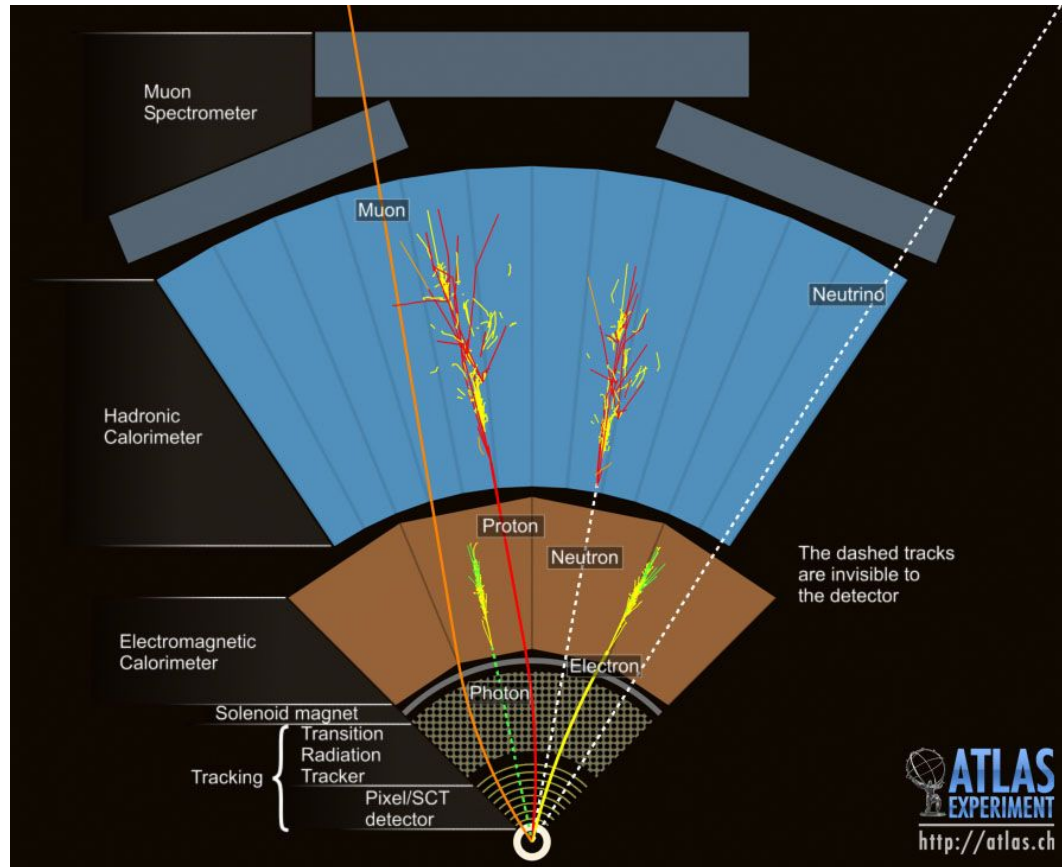
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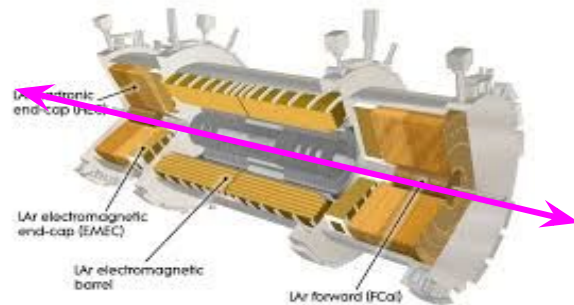
Backup



Non-pointing and delayed photon analysis

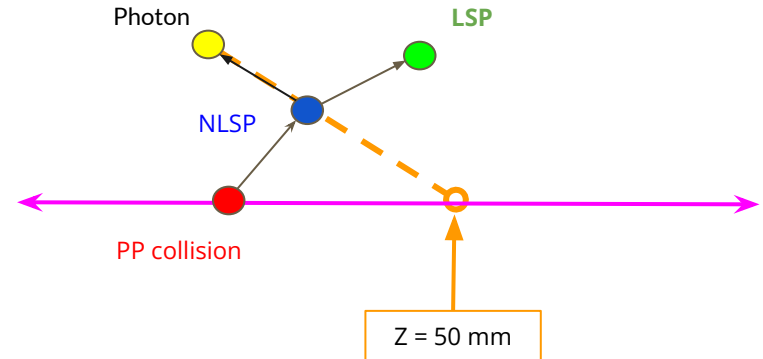
Because of the NLSP's lifetime, the photon it produces will have several distinct properties which could be experimentally detected.

- The photon will be **delayed** with respect to the original pp collision by the non-prompt decay of the NLSP.
- The photon will be **non-pointing** with respect to the primary vertex because it is emitted from a secondary vertex when the NLSP decays.
- The **missing transverse energy (MET)** can be calculated from a summation of the transverse energy from each event → this is the energy carried away by the LSP which remains otherwise invisible to the detectors.



Img credit: CERN

Z axis



Significance in Signal to Background

- Statistical analysis was done for two cases
 - Inclusive channels (1lp_1ph, 2lp_1ph, 1lp_2ph, 2lp_2ph)
 - Two photon channels (1lp_2ph, 2lp_2ph)

- Our MET cuts are split into 7 candidate signal regions extending out to 150 GeV

MET Region
S0 (> 30 GeV)
S1 (> 40 GeV)
S2 (> 50 GeV)
S3 (> 70 GeV)
S4 (> 90 GeV)
S5 (> 110 GeV)
S6 (> 130 GeV)
R9 (> 150 GeV)

- General asymptotic formula for discovery significance is only true for small uncertainty background and signal shapes-- in the high pointing the timing bins, this is not the case. The modified formula includes additional 'shape' terms in quadrature based on their statistical errors

General Formula

$$Z_{ij} = \sqrt{2 \left((S_{ij} + B_{ij}) \log \left(1 + \frac{S_{ij}}{B_{ij}} \right) - S_{ij} \right)}$$

Modified Formula

$$Z'_{ij} = \left(\frac{-1}{2 \left(B_{ij} \log \left(1 + \frac{S_{ij}}{B_{ij}} \right) - S_{ij} \right)} + \frac{B_{ij}}{S_{ij}^2} + \frac{1}{S_{ij}} \right)^{-1/2}$$